

Raghavan Balanathan

+1 (765) 734-7606 | raghavanb2103@gmail.com | West Lafayette, IN | linkedin.com/in/raghavanbala/

Computer Science graduate student seeking Summer 2026 internship to apply Machine Learning and software development skills in building impactful applications. **U.S. citizen** with full work authorization.

EDUCATION

Purdue University

Master of Science in Computer Science

Courses Enrolled: Computer Networks, Databases, Statistical Machine Learning

West Lafayette, IN, USA

May 2027 (Expected)

National Institute of Technology

Bachelor of Technology(Honors) in Computer Science and Engineering

GPA: 3.88/4.0 | Rank: 1/141

Tiruchirappalli, India

May 2025

EXPERIENCE

Spring Lab-Indian Institute of Technology Madras

Research Intern

Chennai, India

Dec 2023 – Apr 2025

- Achieved **<10%** Word Error Rate (WER) using **self-supervised models** like HuBERT and LLMs to perform **multi-modal multi-tasking** with discrete speech units for Automatic Speech Recognition.
- Performed **continual pretraining** and **fine-tuning** of Large Language Models (LLMs) such as Llama, GPT using **QLoRA** within Fully Sharded Data Parallel (FSDP) settings, processing **~1 TB** of speech and text data to handle Big Data-scale training.
- Developed transformer-based query-by-example spoken keyword detection model (E2E-QBE-SKD) with 0.8 MTWV on 35.7K speech keywords, demonstrating end-to-end pipeline development.

PROJECTS

Continual Learning for Video Segmentation of Surgical Tool

B.Tech Thesis 📄

Tiruchirappalli, India

Jan 2025 – May 2025

- Developed a **Domain-Incremental Continual Learning** framework for surgical tool segmentation using **SAM2** trained with **LoRA**, enabling privacy-preserving adaptation to challenging surgical scenes.
- Improved segmentation accuracy by **up to 5%** using CLIP with K-Means parameter selection, outperforming LwF baselines.

SummarizeIQ: Article Summarizer

Schlumberger's Hackathon, IIT Madras 📄

Chennai, India

Jan 2023 – Feb 2023

- Engineered a **FastAPI microservice** using **TorchServe** to run the **T5 Transformer model**, providing scalable, low-latency abstractive NLP summarization of webpage articles.
- Implemented **LexRank** and **TextRank** algorithms within API pipelines for **extractive summarization**, optimizing asynchronous request handling and parallel scraping for efficient data processing.
- Delivered a scalable summarization dashboard, securing **4th place** out of **1,150 teams**

PUBLICATIONS

- MICCAI Workshop on Medical AI 2025:** Gokul Adethya, Nitish N, **Raghavan Balanathan**, and K Sitara, "Domain-Incremental Continual Learning for Robust Surgical Tool Segmentation". (Accepted) 📄

TECHNICAL SKILLS

- Languages:** Python, C++, C, JavaScript, React, SQL
- Libraries/Tools:** NumPy, pandas, scikit-learn, Matplotlib, PyTorch, TensorFlow, Git, Docker.
- Areas of Interest:** Machine Learning, Natural Language Processing, Multimodality, Data Science